QED3 REVISIONS: 2026-10-23

Abstract. Minor corrections and local clarifications to the comprehensive QED3 manuscript and the corresponding companion passages, dated 23 October 2026. The main conclusions are unchanged. The local fold-proof correction restores the supplied argument already present in the reflection-positivity companion; no new construction or estimate is introduced.

Contents

1. Corrections 1

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The following list covers audit items QED-0001–QED-0042. Corrections already present in a companion were retained. Matching TeX sources and PDFs were refreshed in the repository and Dropbox collaboration copy.

1. Clarified the abstract's weak-charge and ergodic-component scope, the lattice-unit side length, charge-uniformity of the cumulant remainder, and the one-parameter translation group used in the clustering proof (1–4).
2. Allowed the notation table to break between rows, kept the two-line continuity estimate together, and replaced inaccurate "first-page statement" references by theorem references (5, 34, 42).
3. Specified Euclidean face area, described the hopping map as a phase-weighted site permutation, and stated the bound for each Grassmann coefficient (6, 7, 14).
4. Made spelling consistent within each manuscript; corrected punctuation and subject–verb agreement (8, 32, 37).
5. Assigned upper-slab coefficient support to S_+ and explicitly retained the generator support and coefficient-variable dependence in $\mathcal{G}^+$ (9, 10).
6. Identified the integer Fourier coefficients in the cut-link integral, compared Berezin integration functionals consistently, and defined the normalized Gaussian at first use (11, 12, 16).
7. Based each folded cyclic trace at its own slab boundary, including half-integer-width applications. The trace identity and chessboard estimate are unchanged (13).
8. Fixed the dual-tree root, removed an undefined cutoff exponent, and specified the unit-lattice block and its exact references (15, 17, 26).
9. Distinguished analytic bilinear pairings from auxiliary Hermitian Fourier products, made the conjugated connection explicit, and separated operator and trace norms from weighted block norms (18, 29, 36).
10. Distinguished thick layers from chambers, photon overlap components from fermion face components, and localization-family indices from torus indices. Stated the common incoming data required in primitive comparisons (19, 30, 31, 33).
11. Retained the scalar coefficient of a constant polynomial, restored the factor $2\pi/q$ in the local lift, and specified unit mirror normals (20–22).

12. Corrected the Bochner-integrability explanation, displayed the spectral projections in the finite-stage identity, and used absolute coordinate differences for the positive displacement (23–25).
13. Named the chosen real summand rather than the full real part. Renamed the scalar polymer-sum constant K_{pol}, supplied its definition and reference, and clarified the individual counterterm bounds before summation; operator and compact-set uses of K_0 remain unchanged (27, 28).
14. Restricted cutoff independence to the stop offset and physical stop quantities, specified nonnegative Gaussian-tail thresholds, distinguished basis momenta from twist-shifted symbol arguments, restored the mean-zero transverse field domain, and cited the selected coefficient subsequence (35, 38–41).

The refreshed companion exports also retain the project's current muted link colors. Original source inputs and historical copies are unchanged.